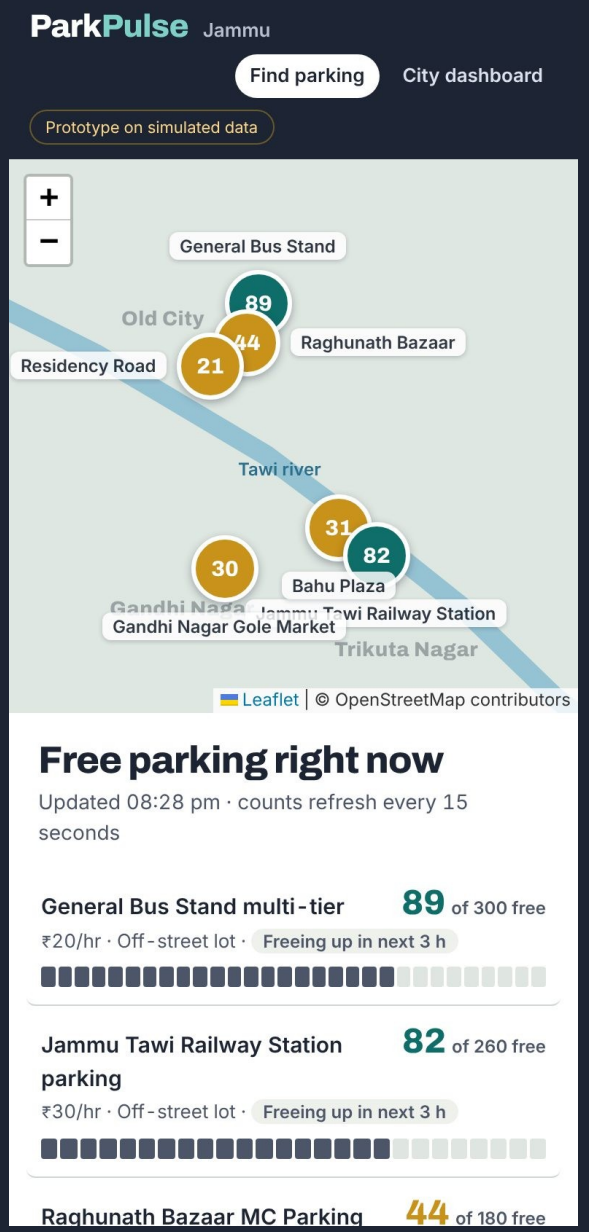
Working prototype on simulated data

ParkPulse Jammu

Live parking and congestion intelligence from the city's existing cameras

Digital Governance, Mobility & Smart Infrastructure

Problem statement: use real-time data and technology to optimise parking, manage congestion and discourage illegal parking.



Jammu's parking problem is an information problem

Bays exist, but nobody can see which ones are free, so drivers circle, double-park and block junctions.

Drivers circle the markets

Raghunath Bazaar, Residency Road and Gandhi Nagar fill at midday and evening. With no live information, drivers search street by street and add to the jam.

Illegal parking blocks traffic

Vehicles left at City Chowk, Gole Market and station roads choke junctions. Enforcement relies on patrols that cannot be everywhere.

Pilgrim and festival surges

Vaishno Devi pilgrims, Navratri and Diwali bring visitors who do not know the city, all arriving at the railway station and bus stand.

The city plans without data

JMC and JSCL have no hour-by-hour view of where and when lots fill, or how much parking fee goes uncollected.

One camera feed, three jobs

Citizens and pilgrims

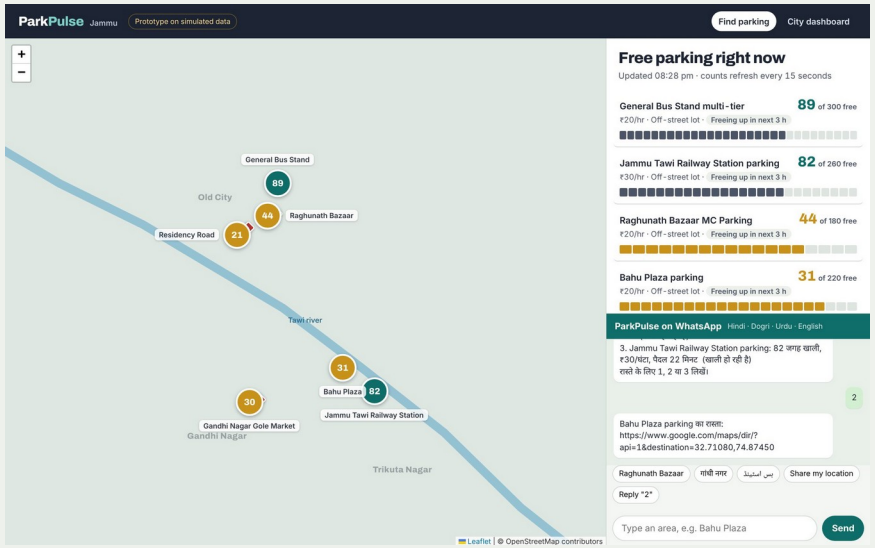
Free-bay counts on a web map and a WhatsApp bot in Hindi, Dogri, Urdu and English. Reply 1, 2 or 3 for directions. No app to install.

Traffic police

An automatic alert when a vehicle stays in a no-parking zone past 5 minutes, with location and vehicle type.

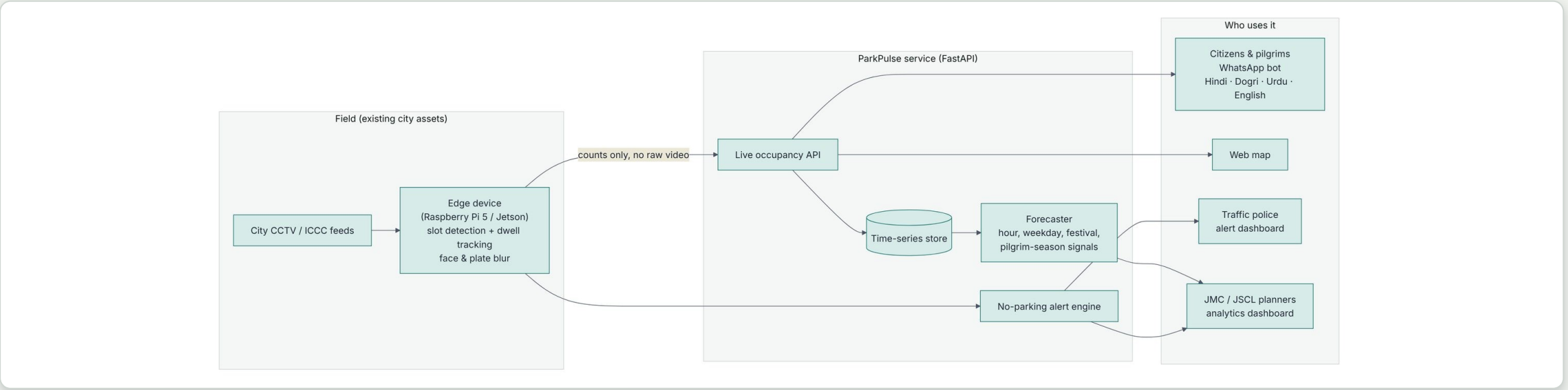
JMC and JSCL planners

Hour-by-hour fill patterns, a 6-hour forecast per zone, enforcement hotspots and an estimate of uncollected fees.



Citizen map, live in the prototype

How it works



Edge, next to the camera

Raspberry Pi 5 or Jetson Orin Nano. OpenCV slot counter by default, optional YOLO for hard angles. Faces and plates blurred on the device; only counts leave it.

City service

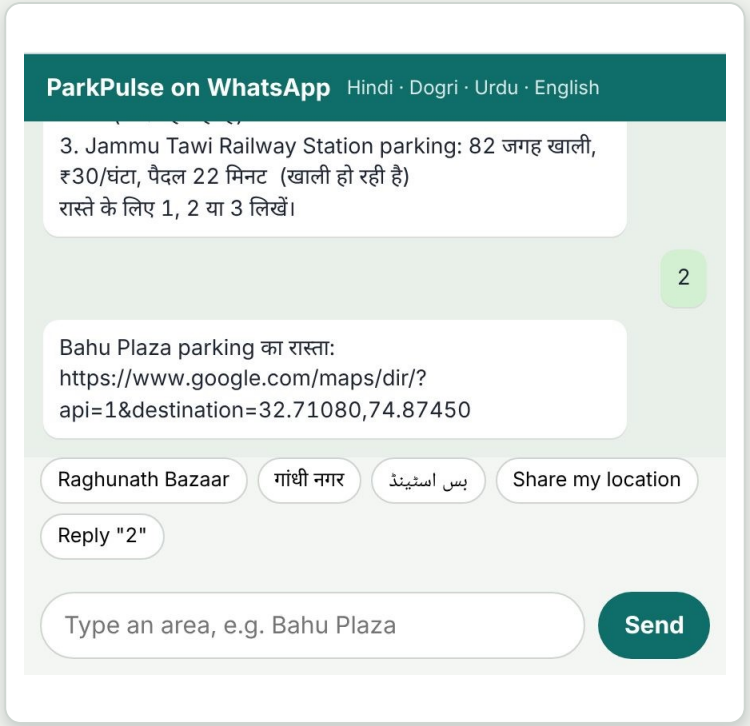
Python and FastAPI keep live occupancy, run per-zone forecasts with gradient boosting and raise alerts. Ships as one Docker image.

People

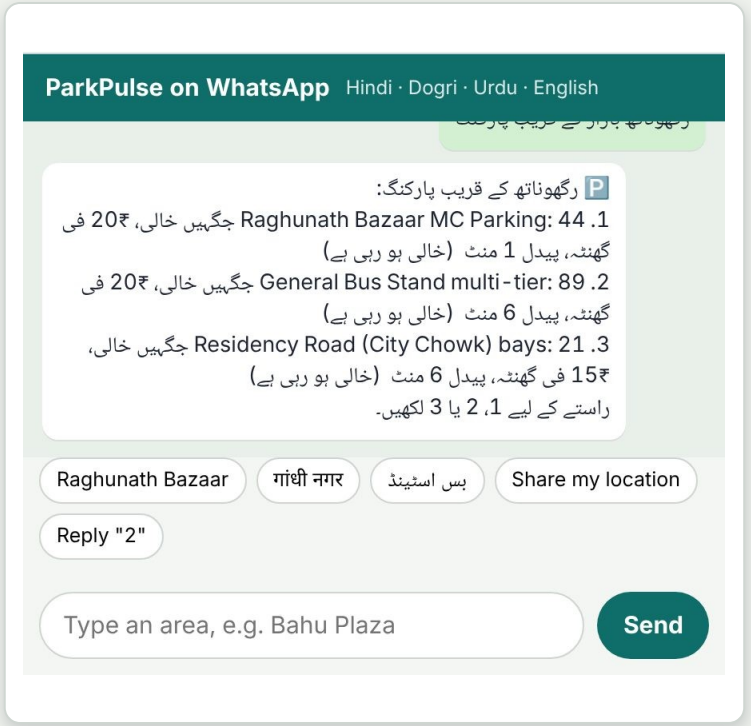
WhatsApp Cloud API webhook, a Leaflet web map for citizens and a dashboard for officials. Works offline for demos.

Ask in your language, get the nearest free bay

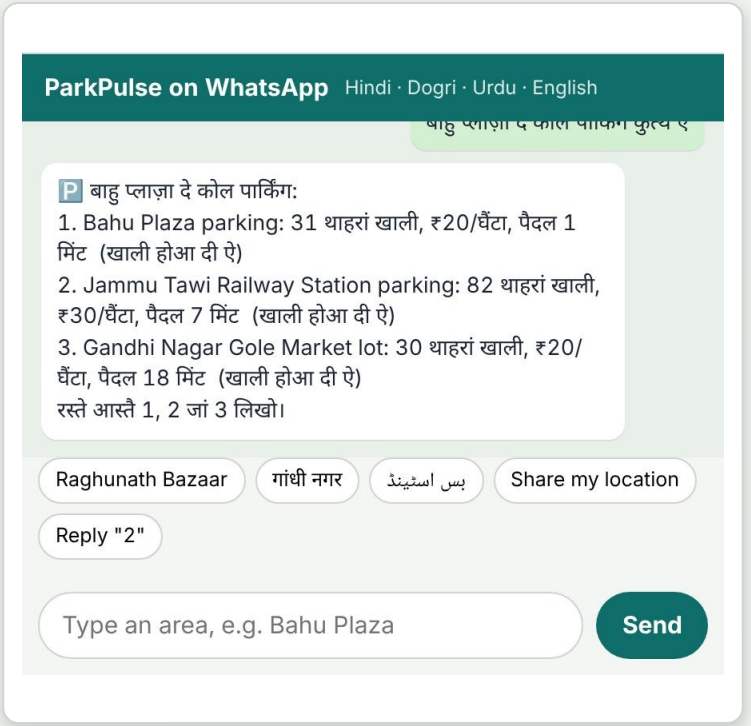
Type an area name (typos are fine) or share a location pin. The bot lists the three nearest zones with free bays, fee and walking time.



Hindi



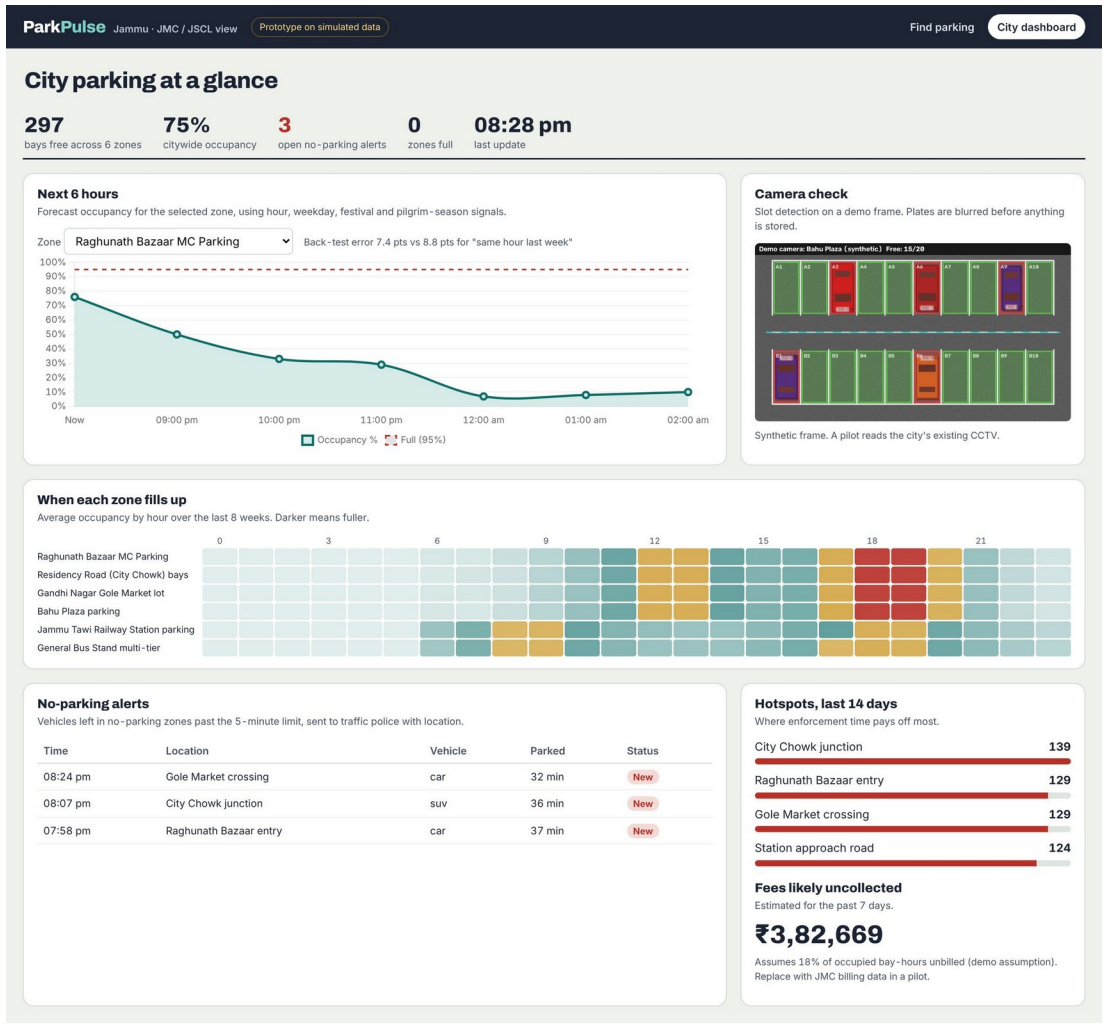
Urdu, right to left



Dogri

Dogri and Urdu replies are a first draft and will be reviewed by native speakers before launch.

A dashboard the city can act on



Heatmap of fill by hour

Market zones fill at 12–13h and 18–19h; the bus stand and station peak at 8–9h. That is when marshals and overflow lots help most.

6-hour forecast

Warns before a zone crosses 95% full, so fees or marshals can be adjusted hours ahead.

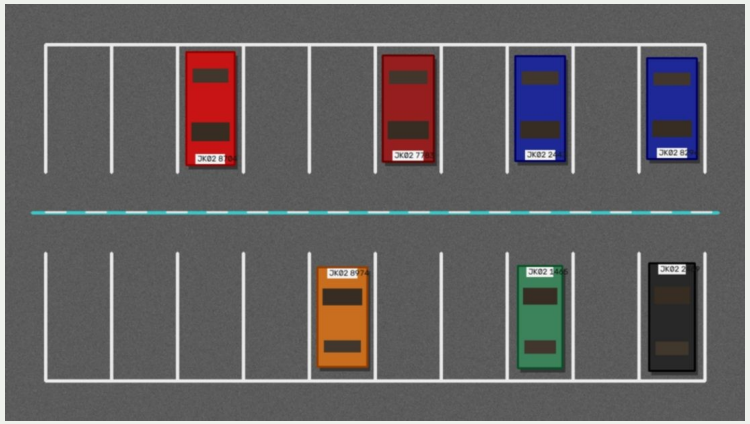
Alerts and hotspots

Live no-parking alerts, and segments ranked by alerts over 14 days so patrols go where they matter.

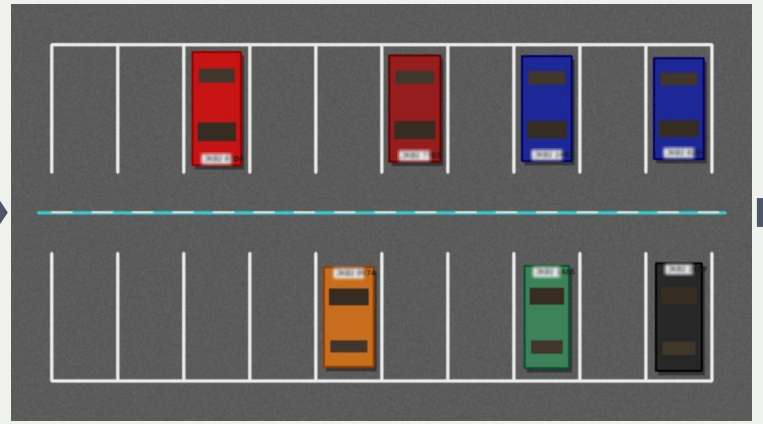
Uncollected fees

An estimate of fees lost to unbilled and overstayed parking, to size the revenue case.

From camera frame to free-bay count, privately



1. Camera frame



2. Plates and faces blurred



3. Each bay marked free or taken

100%

bays classified correctly

89%

number plates found and blurred

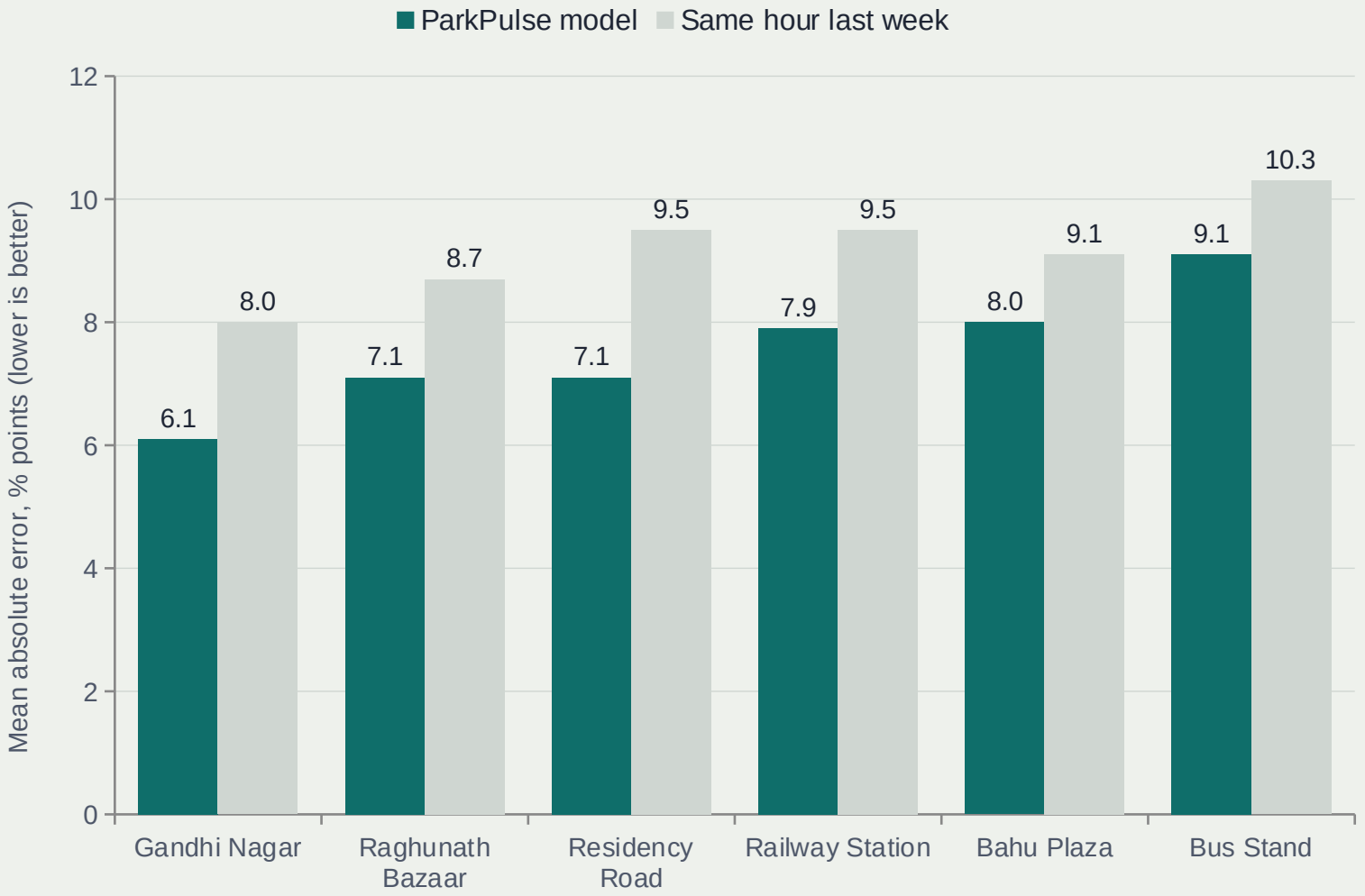
1 alert

per vehicle after a 5-minute no-parking stay

Measured on 50 synthetic frames (1,000 bays).

Synthetic frames are easy. On Jammu CCTV we draw bay outlines per camera with the included tool, switch oblique cameras to YOLO, and use a trained Indian-plate detector before any deployment.

Forecasts beat the simple rule in all six zones



1.1–2.4

percentage points less error than the simple rule, depending on zone

Signals the model uses

- Hour and weekday
- Festival windows (Navratri, Diwali)
- Pilgrim-season index
- Same hour yesterday and last week

Back-tested on the last 7 of 8 weeks of simulated history. Real error is measured in the pilot.

What makes ParkPulse different

Reuses the city's cameras

Almost no new hardware, so a pilot starts in weeks, not months.

WhatsApp, in local languages

Reaches residents and pilgrims who will never install another app.

Three jobs from one system

Guides drivers, feeds enforcement and gives planners data at the same time.

Tuned to Jammu

Forecasts know market days, festivals and the Vaishno Devi pilgrim season.

Private by default

Blurring on the device and counts-only uploads make it easier for a government body to adopt.

Cheap to run

Classical vision runs on a Raspberry Pi; open-source stack, no licence fees.

Impact we will measure in a 12-week pilot

Targets for 2–3 high-demand zones, measured against a pre-pilot baseline. These are goals, not results.

25–30%

less time spent searching for parking

20–25%

fewer illegal-parking incidents

15–20%

more parking fee collected

Also expected

- Less circling in markets, so less congestion and fewer emissions
- Steadier footfall for shopkeepers; an easier first visit for pilgrims and tourists
- Data for JMC and JSCL to set capacity, pricing and enforcement ward by ward

A single-zone pilot costs under ₹1 lakh

Item	Approx. cost
2 edge AI devices (Raspberry Pi 5 + accelerator, or Jetson Orin Nano)	₹25,000–45,000
2–3 IP cameras with mounts (only if city feeds are not shared)	₹10,000–18,000
4G routers and data SIMs	₹5,000–8,000
Cloud hosting, database and maps, 3 months	₹6,000–12,000
WhatsApp Business API messaging	₹3,000–6,000
Power, weatherproofing, testing	₹5,000–10,000
Total	**₹55,000–1,00,000**

Prices are typical Indian retail estimates.

≈ ₹75,000

for one zone with our own cameras

≈ ₹20,000

if the city shares existing CCTV feeds

The software is already built and open source, so the money goes on hardware and running costs.

12 weeks from approval to a rollout decision

Weeks 1–2

Pick zones

Choose 2–3 zones with JMC, such as Raghunath Bazaar, Residency Road and Gandhi Nagar. Get camera access or mount low-cost cameras.

Weeks 3–4

Calibrate

Draw bay outlines, label a few hundred real frames, pick the vision backend per camera and check accuracy against manual counts.

Weeks 5–8

Go live

Open the bot to a test group, route alerts to one traffic police unit, review Dogri and Urdu copy with native speakers.

Weeks 9–12

Measure

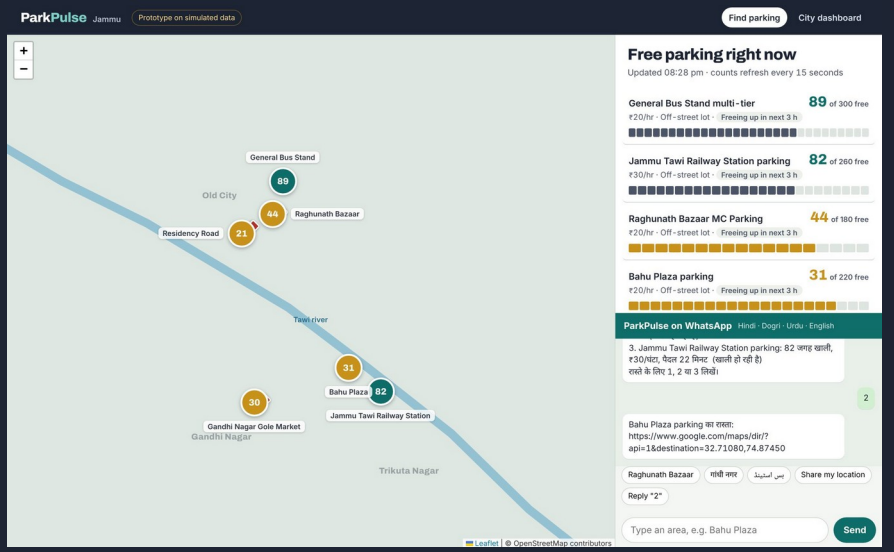
Compare search time, illegal parking and fee collection with the baseline, then decide on ward-by-ward rollout.

What we need from the city

Read access to camera feeds in the pilot zones, current parking capacity and fee records, a traffic police contact for alerts, and mentorship from JSCL's command centre team.

Built, running, ready for Jammu's cameras

Every screen in this deck runs today. Connect two or three city camera feeds and we can measure real impact within a quarter.



Code, tests and docs

github.com/Aditya-eng/ParkPulse

Team Byte Bros